Ohm's Law: $R = \frac{v}{i}$, $v = iR$, $i = \frac{v}{R}$ $\quad (\Omega = \frac{V}{A})$

conductance: $G = \frac{1}{R} = \frac{i}{v}$ $\quad (\mho = S)$

parallel resistors: $R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}\right)^{-1}$

series resistors: $R_{eq} = R_1 + R_2 + \dots + R_N$

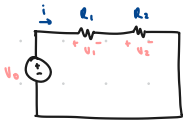
parallel conductances: $G_{eq} = G_1 + G_2 + \dots + G_N$

series conductances: $G_{eq} = \left(\frac{1}{G_1} + \frac{1}{G_2} + \dots + \frac{1}{G_N}\right)^{-1}$

Kirchoff's current law: current entering node = current exiting node

Kirchoff's voltage law: voltage rises around loop = voltage drops around loop

Voltage division: series resistors

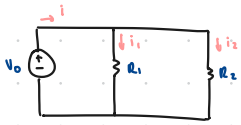


$v_1 = v_0\left(\frac{R_1}{R_1+R_2}\right)$

$v_2 = v_0\left(\frac{R_2}{R_1+R_2}\right)$

$v_N = v_0\left(\frac{R_N}{R_1+R_2+\dots+R_N}\right)$

Current division: parallel resistors



$i_1 = i_0\frac{R_2}{R_1+R_2}$

$i_2 = i_0\frac{R_1}{R_1+R_2}$

$i_N = i_0\left(\frac{G_N}{G_1+G_2+\dots+G_N}\right)$

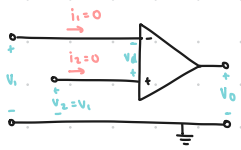
Power: $P = vi$, +ve if absorb (higher → lower voltage)

−ve if deliver (lower → higher voltage)

Operational amplifiers: acts as voltage amplifier

large input resistance, small output resistance, large voltage gain

ideal op-amp model:



① $R_{in} = \infty$

② $R_{out} = 0$

③ $A_0 = \infty$ (voltage gain)

④ $i_+ = i_- = 0$ (zero input current)

⑤ $v_d = v_{in} = v_+ - v_- = 0$ since $v_+ = v_-$

note: never apply KCL @ ground node of op-amp

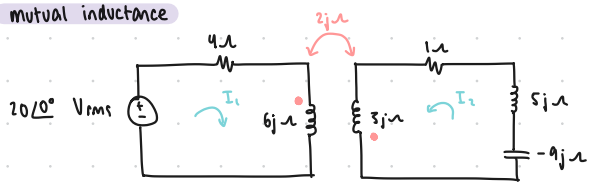
saturation: output of op-amp is limited to DC supply voltage

positive: $v_0 = +v_{cc}$

linear: $-v_{cc} \le v_0 = Av_{in} \le v_{cc}$

negative: $v_0 = -v_{cc}$

mutual inductance



$$\begin{cases} -20\angle 0^\circ + (4+6j)I_1 - 2jI_2 = 0 \\ (-4j+5j+1+3j)I_2 - 2jI_1 = 0 \end{cases}$$

$$\begin{bmatrix} 4+6j & -2j \\ -2j & 1-j \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 20\angle 0^\circ \\ 0 \end{bmatrix}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{\det\begin{bmatrix} 20\angle 0^\circ & -2j \\ 0 & 1-j \end{bmatrix}}{\det\begin{bmatrix} 4+6j & -2j \\ -2j & 1-j \end{bmatrix}}$$

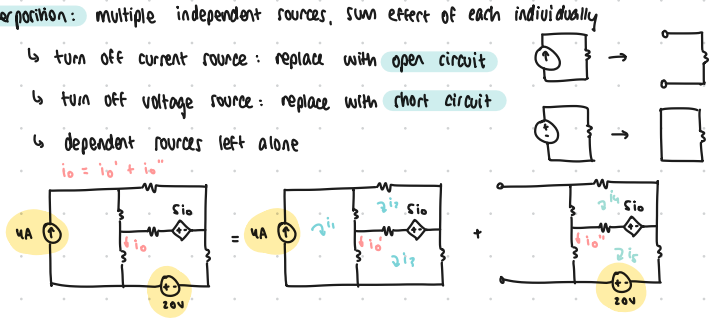
Linear circuit: scaling and additivity

superposition: multiple independent sources, sum effect of each individually

↳ turn off current source: replace with open circuit

↳ turn off voltage source: replace with short circuit

↳ dependent sources left alone



source transformation: simplify voltage source in series or current source in parallel



$v_s = i_s R$

$i_s = \frac{v_s}{R}$

no nodes attached to other circuit elements can be "lost"

Thevenin and Norton's Equivalents

Equivalent circuit has same current and voltage at output terminals



$R_{eq} = R_{Th} = R_N = \frac{V_{oc}}{i_{sc}}$

R_{Th}:

① only independent sources: short all voltage sources, open all current sources

② only dependent sources: test voltage (1V) or test charge (1A) to output terminals with node removed → equivalent circuit is just $R_{Th} = \frac{V_{oc}}{i_{sc}}$

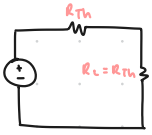
③ both independent and dependent sources: determine both V_{oc} and i_{sc}, $R_{Th} = \frac{V_{oc}}{i_{sc}}$

V_{oc} or I_{sc}:

① Remove load and replace with open circuit for V_{oc} (Thevenin) or short circuit for I_{sc} (Norton)

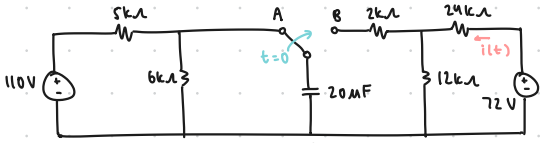
Then, reconnect load with equivalent Thevenin/Norton which replicates original.

Maximum power transfer: $R_L = R_{Th}$

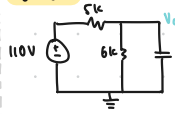


$P_L = P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{I_N^2 R_{Th}}{4}$

transient circuit



$t < 0$:

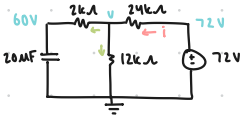


$v_c = (110V)\left(\frac{6}{6+5}\right) = 60V$

$W = \frac{1}{2}CV^2 = \frac{1}{2}(20mF)(60V)^2 = 0.036J$

$t > 0$:

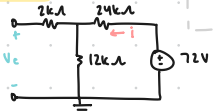
$v_c(0^+) = v_c(0^-) = 60V$



$\frac{72-v}{24} = \frac{v-60}{2} + \frac{v}{12}$

$i(0^+) = \frac{72-v}{24} = 0.8mA$

$t \to \infty$:



$-72 + (24+12)i = 0 \rightarrow i(\infty) = 2mA$

$v_c(\infty) = 72 - 24i \rightarrow v(\infty) = 24V$

$R_{Th} = 2 + 12\|24 = 10k\Omega$

$\tau = R_{Th}C = 0.2s$

$$i(t) = 2 + (0.8-2)e^{-5t}$$

$$v(t) = 24 + (60-24)e^{-5t}$$

Made with Goodnotes

Capacitors

capacitance = charges on electrodes relative to potential difference

$C = \frac{Q}{V}$ → $Q = CV$ [F = C/V]

$W = \frac{1}{2}CV_c^2 = \frac{Q_c^2}{2C}$

parallel: $C_T = C_1 + C_2 + \dots + C_N$

series: $C_T = \left(\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}\right)^{-1}$

Inductors

inductance = magnetic flux relative to current used to create it

$L = \frac{\text{total magnetic flux}}{\text{source of flux}} = \frac{N\Phi}{i}$ [$H = \frac{Wb}{A}$]

series: $L_T = L_1 + L_2 + \dots + L_n$

parallel: $L_T = \left(\frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_n}\right)^{-1}$

energy: $W = \frac{1}{2}Li^2$

ideal solenoidal inductor: $L = \frac{N^2 \mu_r \mu_0 A}{l}$

magnetic field: $B = \frac{\mu_r \mu_0 N i}{l}$

Mutual Inductance

Ratio of applied current to mutual flux with second coil

$V_2 = M_{21}\frac{di_1}{dt}$ $V_1 = M_{12}\frac{di_2}{dt}$ $M_{12} = M_{21} = M$

$M = k\sqrt{L_1 L_2}$ (k: coupling coefficient)

Dot convention:

- if current enters dot in one coil, polarity +ve at dot in second
- if current leaves dot in one coil, polarity -ve at dot in second

Ideal Transformers

create induced voltage on secondary winding

$\frac{V_2}{V_1} = \pm n = \frac{N_2}{N_1}$ $\frac{I_2}{I_1} = \mp\frac{1}{n} = \mp\frac{N_1}{N_2}$ $n = \frac{N_2}{N_1}$

n>1: step-up, n<1: step-down, n=1: isolation

At dotted terminals:

voltage both pos. / both neg.: $\frac{V_2}{V_1} = +n$, otherwise $\frac{V_2}{V_1} = -n$

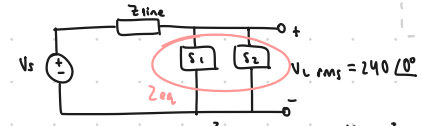
currents both enter / both leave: $\frac{I_2}{I_1} = -\frac{1}{n}$, otherwise $\frac{I_2}{I_1} = +\frac{1}{n}$

Transform a load impedance: $Z_{in} = \frac{Z_L}{n^2}$ "match" load to source for max

Complex power is conserved: $S_1 = V_1 I_1^* = V_2 I_2^* = S_2$

Loads and sources

$f = 60Hz \Rightarrow \omega = 120\pi$



$V_{L\,rms} = 240\angle 0°$

$Z_{L1}^* = \frac{V_{L\,rms}^2}{S_1}$, $Z_{L2}^* = \frac{V_{L\,rms}^2}{S_2}$

$V_L = (V_s)\frac{Z_{eq}}{Z_{eq} + Z_{line}}$

suppose $S_1 = 60 + j37.18$ kVA

$S_2 = 40 + j22.09$ kVA

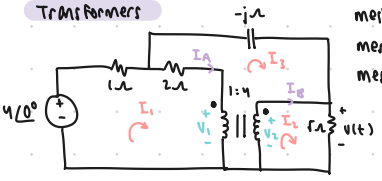
$\therefore P = 100$ kW

$Z_{eq} = 0.472\angle 34.4°$

add capacitor to correct to unity (0°):

$C = \frac{100\text{kW}(\tan 34.4° - \tan 0°)}{(120\pi)(240\angle 0°)^2}$

Transformers



mesh #1: $-4 + I_1 + 2(I_1 - I_3) + V_1 = 0$

mesh #2: $5I_2 - V_2 = 0$

mesh #3: $-jI_3 + V_2 - V_1 + 2(I_3 - I_1) = 0$

also: $V_2 = 4V_1$ (or $I_B = \frac{1}{4}I_A$)

First-order circuits

IF: $v' + P(t)v = f(t)$, $IF = \rho(t) = e^{\int P(t)dt}$ $\Rightarrow v(t) = \frac{1}{\rho(t)}\left[\int \rho(t) f(t)\,dt + C\right]$

Inductors: $V_L(t) = L\frac{di_L(t)}{dt}$, $i_L(0^-) = i_L(0^+)$

current cannot change instantaneously, looks like short circuit

Capacitors: $i_C(t) = C\frac{dV_C(t)}{dt}$, $V_C(0^-) = V_C(0^+)$

voltage cannot change instantaneously, looks like open circuit

source free RC/RL circuit: $v(t) = v(0^+)e^{-t/\tau}$, $i(t) = i(0^+)e^{-t/\tau}$

$\tau = R_{Th}C = \frac{L}{R_{Th}}$ (time constant) R_{Th} is "seen" by capacitor/inductor

circuit with source (steady state + transient):

$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$

$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$

(natural: $v(0) - v(\infty)$, $i(0) - i(\infty)$; forced: $v(\infty)$, $i(\infty)$)

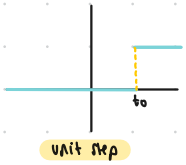
① $t < 0$: determine initial values

② $t = 0^+$: replace inductor with current source, capacitor with voltage source

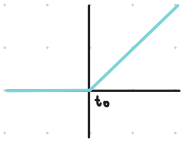
solve for natural response (inductor/capacitor) and forced response (original)

③ $t > 0$: find R_{Th} seen by inductor/capacitor

④ $t \to \infty$: $i(\infty)$ and $v(\infty)$



unit step



ramp



delta

$u(t-t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$ $r(t-t_0) = \begin{cases} 0, & t < t_0 \\ t - t_0, & t > t_0 \end{cases}$ $\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{undef}, & t = 0 \\ 0, & t > 0 \end{cases}$

Complex power

$Z = \frac{1}{j\omega C} = j\omega L$ (resistance / reactance)

"R" Impedence: $Z = \frac{V}{I} = R(\omega) + jX(\omega)$ (Ω)

"G" Admittance: $Y = \frac{1}{Z} = \frac{I}{V} = G(\omega) + jB(\omega)$ (S) (conductance / susceptance)

resistors: real $Z_R = \frac{V}{I}\angle 0°$

capacitors: imaginary, $X < 0$ $Z_C = \frac{V}{I}\angle -90°$

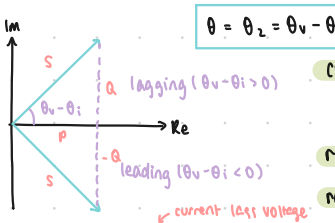
inductors: imaginary, $X > 0$ $Z_L = \frac{V}{I}\angle 90°$

$\omega = 2\pi f$

$v(t) = V_m\cos(\omega t + \theta_v)$ → $V = V_{rms}\angle\theta_v$ $V_{rms} = \frac{V_m}{\sqrt{2}}$ ($\frac{V_m}{\sqrt{2}}\angle\theta_v$ phasor)

$i(t) = I_m\cos(\omega t + \theta_i)$ → $I = I_{rms}\angle\theta_i$ $I_{rms} = \frac{I_m}{\sqrt{2}}$ ($\frac{I_m}{\sqrt{2}}\angle\theta_i$ phasor)

$\theta = \theta_z = \theta_v - \theta_i$



Q > 0: inductive (lagging pf)

Q < 0: capacitive (leading pf)

Q = 0: resistive (unity pf)

complex power: $S = V_{rms}I_{rms}^* = P + jQ$ (P: resistors, real; Q: capacitors/inductors, reactive)

$= \frac{|V_{rms}|^2}{Z^*} = |I_{rms}|^2 Z = \frac{1}{2}VI^*$ (VA)

real (average) power: $P = V_{rms}I_{rms}\cos\theta$ (W)

reactive power: $Q = V_{rms}I_{rms}\sin\theta$ (VAR)

apparent power: $|S| = V_{rms}I_{rms} = \sqrt{P^2 + Q^2}$ (VA)

power factor: $pf = \cos\theta = \frac{P}{|S|}$

power factor correction: $C = \frac{Q_c}{\omega V_{rms}^2} = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V_{rms}^2}$ (original, new; add parallel capacitor)

If $x(t) = A + B\cos(\omega t + \theta_x)$, then $X_{rms} = \sqrt{A^2 + \frac{B^2}{2}}$

root mean square: $X_{rms} = \sqrt{\frac{1}{T}\int_0^T x^2(t)}$ (for non-sinusoidal)

max power transfer

$Z_L = Z_{Th}^* = R_{Th} - jX_{Th}$, $P_{max} = \frac{|V_{Th}|^2}{8R_{Th}}$

Made with Goodnotes